2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.204 Software Design

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Explain how explicitly designing and documenting a software architecture might support large-scale software reuse.

[**2 marks**]

1.b Use a diagram to illustrate the Model View Controller (MVC) architecture style. Identify one disadvantages of the MVC architecture style.

[**5 marks**]

1.c A store requires a software system for managing its laptop rentals. The system should allow customers to rent only one laptop at a time. Figure 1 shows the initial object-oriented design for the system.

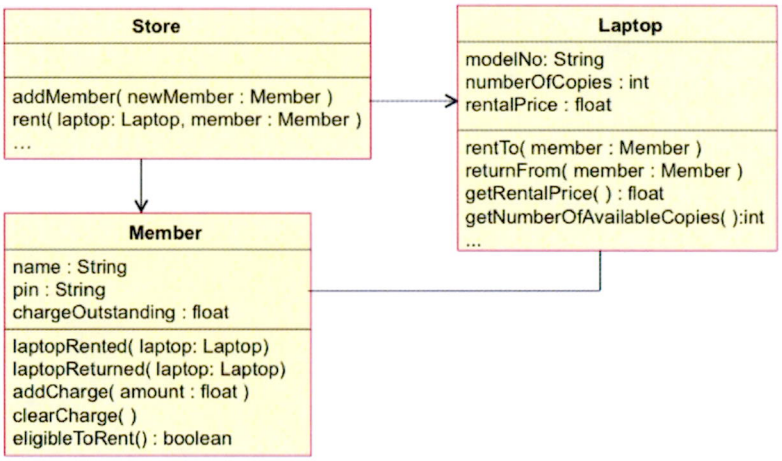


Figure 1. Initial object-oriented design

(i) Give three reasons why this design is not good for reuse

[**3 marks**]

(ii) For a design to be flexible it should anticipate adding new functions and changing existing functions. Modify the rental application design in Figure 1 to:

- Support rentals of WiFi routers and printers, in addition to laptops.
- Support purchases as well as rentals.
- Allow members to rent multiple items simultaneously.

[**9 mark**]

Question 1 continues on next page...

Question 1 continued…

1.d Consider a software system for printing insurance policies. The system accepts XML messages from agency management systems. Incoming messages are based on the ASSURE XML specification, an insurance industry standard. However, each agency has added proprietary extensions to the standard ASSURE transactions. A print request message specifies the type of document to be generated, for example, an HTML document or a Portable Document Format (PDF) document. The request also includes policy data such as client information, coverage, and endorsements. The software system processes the proprietary extensions and adds the jurisdiction-specific information that should appear on the printed documents, such as local or regional requirements and restrictions. The system then generates the documents in the requested format and returns them to the agency management system.

(i) Design an architecture for the printing system based on an appropriate style. Your design should identify the print system's main sub-systems and the data flows between them. You may use box and line diagrams.

[**6 marks**]

Total 25 marks

Question 2

2.a Describe 4 key objectives of software design.

[4 marks]

2.b A hospital intensive care unit is designed to monitor the blood pressure, heart rate and respiration rate of a number of patients. Each patient is in a bed, which has its own individual set of monitoring equipment and an alarm. Each bed has between one and three monitors, each of which can measure one of the three vital signs described. The monitoring equipment is connected to a central management system, which provides a display showing the current values of the vital signs of any patient. The management system also maintains a record of the emergency calls received from each bed. When the monitoring equipment detects an abnormal change in a patient's state, it sets off the alarm and communicates the state of the patient to the management system, which updates its display and pages the relevant nurse. After dealing with the emergency, the nurse can reset the monitoring equipment ready for another emergency.

(i) Identify a set of object classes for the system together with their attributes and possible operations. Show the relationships between the object classes, including inheritance and aggregation.

[**14 marks**]

(ii) Construct a sequence (interaction) diagram showing how the system behaves in an emergency situation (i.e. when the monitoring equipment detects an abnormal change in a patient's condition).

[**7 marks**]

Total 25 marks

Question 3

3.a Explain the Adapter design pattern in terms of its 'intent' and 'when to use'. Draw two class diagrams of the Adapter design pattern: one for 'Object adapter' and the other for 'Class adapter'. The class diagrams should show the objects participating in the pattern, including their relationships.

[12 marks]

3.b Read the following use case for the operation of a soft drink vending machine (SVM) and draw a Statechart that specifies the system behaviour. Your specification should include more than two parallel regions.

[13 marks]

Use case name: Buy a Bottle of Soft Drink
Summary: User inserts coins to SVM and gets a bottle of soft drink.
Primary Actor: User
Secondary Actor: None
Precondition: SVM is idle.

Description:
1. User inserts coins to SVM.
2. SVM calculates the total amount of money.
3. SVM activates a product's select button, for which the price is less than the total amount.
4. User selects a product, for which the selection button is activated.
5. SVM dispenses the product.
6. SVM displays the remaining amount of money.
7. User takes the bottle.
8. User presses a Return button.
9. SVM returns the change, if there is any.

Alternatives:
3a.1: SVM deactivate a product's select button, for which the price is greater than the total amount.
4a.1 User selects a product, for which the selection button is NOT activated.
4a.2 SVM displays the price of the product.
8a.1 The total amount is zero. End.

Exceptions:
4b.1: User presses a Return button. Go to 9.

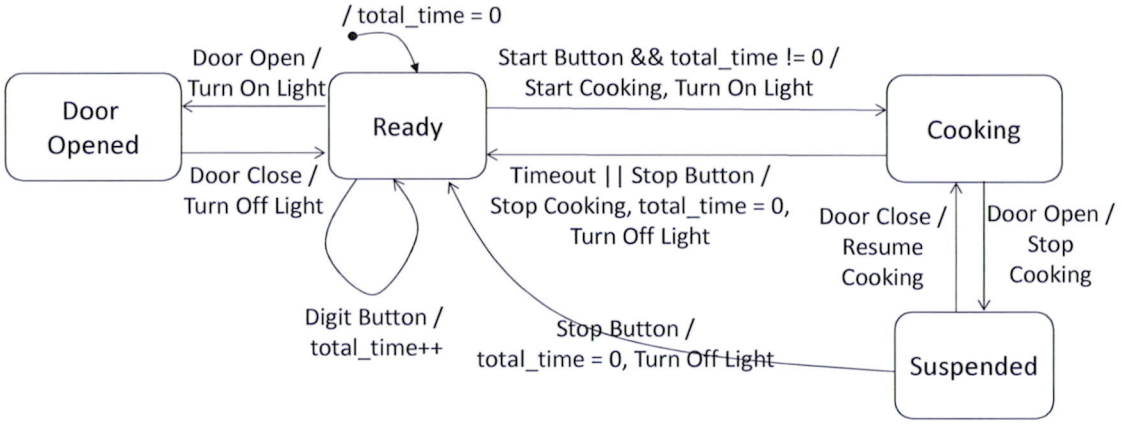
Postcondition: SVM has dispensed a soft drink bottle.

Total 25 marks

Question 4

4.a The State design pattern can be used to define states in a Statechart. Draw a class diagram, which uses the State pattern **to define the states** of the following Statechart for a microwave oven.

[**12 marks**]



4.b Read the following description of a security system for a safe (SSS) and draw a use case diagram of the SSS.

[**13 marks**]

- A security system for a safe (SSS) is connected to a security centre and is equipped with an emergency button and a phone.
- The behaviour of the system changes depending on the current time: daytime or night-time.
- Daytime:
  - It SHALL allow users to use the safe (i.e., open/close) and it SHALL keep a record of its usage.
  - It SHALL allow users to use the phone as a normal landline phone.
  - It SHALL send a notification to the security centre, if the emergency button is pressed.
- Nighttime:
  - It SHALL send a notification to the security centre, if someone opens the safe.
  - It SHALL answer all incoming calls and record any messages.
  - It SHALL send a notification to the security centre, if the emergency button is pressed.

Total 25 marks

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